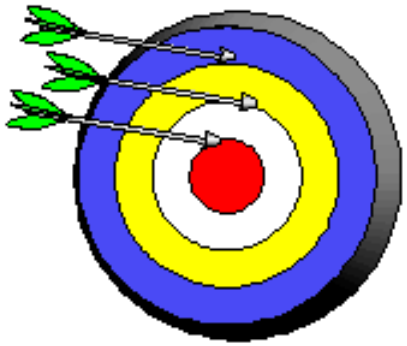
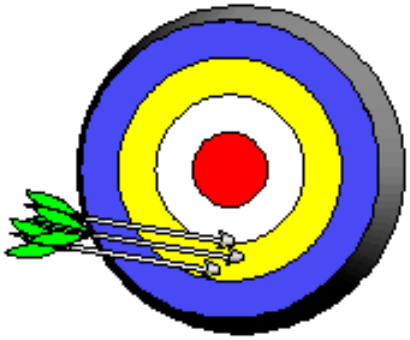
Metrology and Measurement

Precision and Accuracy

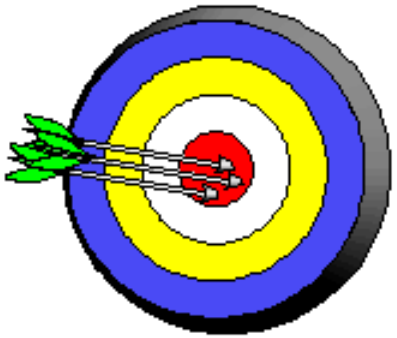
- **Accuracy** refers to the **agreement of a particular value with the true value.**
- **Precision** refers to the **degree of agreement among several measurements** made in the same manner.
  - Degree of exactness, or repeatability/consistency of measurements



Neither accurate nor precise

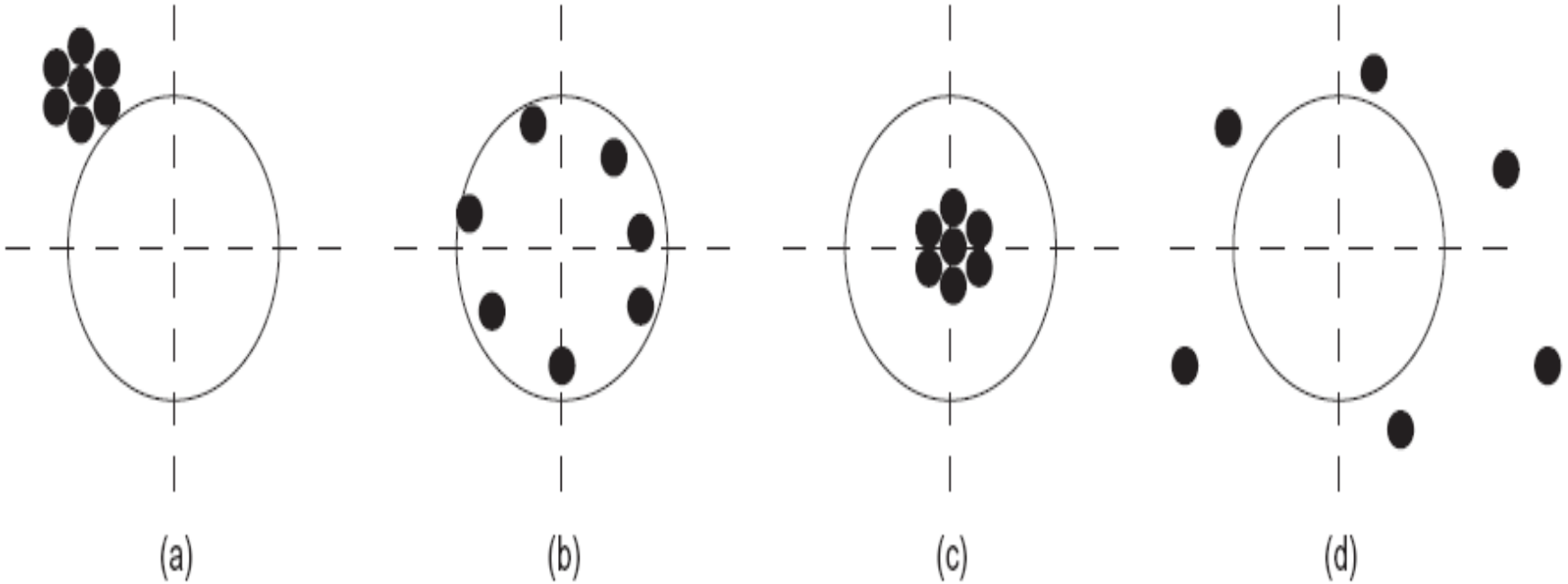


Precise but not accurate



Precise AND accurate

Accuracy vs Precision



Accuracy vs Precision

Volume (mL) of Cough Medicine Delivered by 10-oz (296 mL) Dispensers

Dispenser #1	Dispenser #2	Dispenser #3
283.3	298.3	296.1
284.1	294.2	295.9
283.9	296.0	296.1
284.0	297.8	296.0
284.1	293.9	296.1

Uncertainty in Measurement

- A digit that must be estimated is called uncertain.
- A measurement always has some degree of uncertainty.
- Causes of Uncertainty-
  - Measurements are performed with instruments
  - No instrument can read to an infinite number of decimal places

Uncertainty and Error

- In scientific measurements, **error** refers to the difference between the **measured value** and the **true value** of a quantity.
- It does **not always mean a mistake**, but rather the **uncertainty** or **deviation** inherent in any measurement process.
- Understanding error is crucial in forensic science, where even small inaccuracies can significantly affect case outcomes.

The Error

- The difference between the true value and the mean value of the set of readings on the same component is termed as an *error.*

$$E = V_m - V_t$$

  - where E *is the error,* V_m *the measured value, and* V_t *the true value.*

  - The value of *E is also known as the* ***absolute error.***

Absolute Error

- Absolute error represents the magnitude of the difference between the measured and the true value.
- It indicates how far off a measurement or calculation is from the actual value.
- Absolute error is expressed in the same units as the measurement itself.
  - Absolute Error = |Measured Value−True Value|
    - True weight = 50.00 g, Measured weight = 49.70 g
    - Absolute Error = |49.70−50.00| = 0.30 g

Relative Error

- Relative error expresses the absolute error as a percentage of the true or accepted value.
- It provides a measure of the **relative discrepancy** between the measured and the actual value.
- Relative error is **dimensionless** and is typically expressed as a percentage.

Formula

$$\text{Relative Error} = \frac{\text{Absolute Error}}{\text{True Value}}$$

$$\text{Relative Error (\%)} = \left[\frac{\text{Absolute Error}}{\text{True Value}}\right] \times 100$$

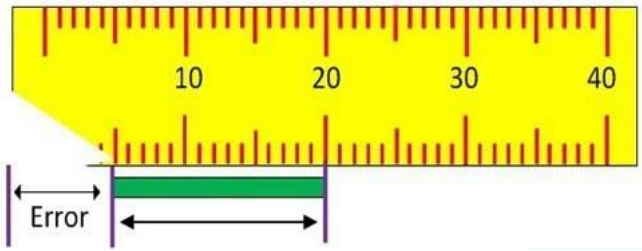
Absolute and Relative Errors

- Absolute error does not provide information about the scale or magnitude of the measured or calculated value; it only measures the discrepancy between the observed and true values.

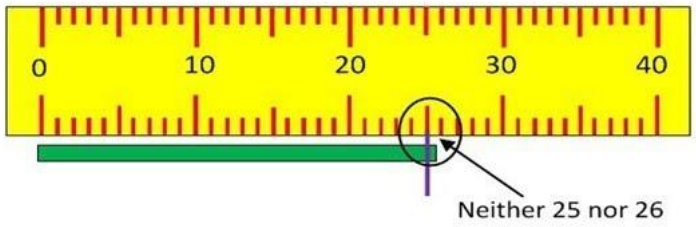
- Relative error allows for comparison of errors across different scales of measurement and provides context for the accuracy of the measurement or calculation.

Measurement Error

Systematic Error



Random Error



Systematic Errors

- Systematic errors are **consistent, repeatable** errors associated with faulty equipment, improper calibration, or flawed methodology.
- These errors cause a bias in the measurements, leading to a consistent deviation from the true value.
- Systematic errors are **due to identified causes** and **can be eliminated**.

- **Characteristics**

1. **Predictable:** Systematic errors occur in a predictable manner, meaning they will consistently affect the measurement in the same direction (either higher or lower than the true value).
2. **Consistent:** These errors affect all measurements in the same way, so they can often be identified and corrected once the source of the error is found.
3. **Affect Accuracy:** Systematic errors primarily affect the **accuracy** of the measurements, as they result in a deviation from the true value.

Systematic Errors

1. **Calibration Error:** If a weighing scale consistently reads 0.05 g higher than the actual value due to improper calibration, all weight measurements will have a systematic error of +0.05 g.
2. **Instrumental Error:** In a forensic investigation, if a gas chromatograph detector consistently reads concentrations higher than the actual levels due to a malfunction, the results will be biased in one direction.

- **Correcting Systematic Errors:**
  - Calibration of instruments.
  - Changing experimental design or method.
  - Applying correction factors to account for known errors.

Random Errors

- Random errors are **unpredictable variations** in the measurement process that occur due to unknown or uncontrollable factors.
- These errors can be **positive or negative**, fluctuating randomly around the true value.
- Due to unknown/uncontrollable origins, it is **not possible to eliminate** these errors.

- **Characteristics**
  - **Unpredictable:** Random errors occur in an unpredictable manner and are different in each measurement. They follow a normal distribution, meaning they can sometimes make the measurement higher or lower than the true value.
  - **Inconsistent:** These errors vary in magnitude and direction from one measurement to the next.
  - **Affect Precision:** Random errors primarily affect the **precision** of the measurements, causing variation or scatter in repeated measurements.

Random Errors

1. **Human Error:** Variations in the way a forensic analyst reads measurements from an instrument (e.g., slight variations in timing during manual tests).
2. **Environmental Fluctuations:** Slight, unpredictable changes in temperature, humidity, or light that affect the performance of sensitive instruments.
3. **Instrument Noise:** In a forensic spectrophotometer, random fluctuations in the output signal caused by electronic noise can result in slightly different absorbance readings for the same sample.

- **Minimizing Random Errors:**
  - Repeating measurements and averaging the results.
  - Improving experimental techniques.
  - Using more precise instruments with less variability.

Gross Errors/Blunders

- The errors, which occur due to the lack of experience of the observer while taking the measurement values are known as **gross errors**.
  - Also known as blunder, or an **outright mistake**.
  - The values of gross errors will vary from observer to observer.

Causes:

1. A person **may record a wrong value**,
2. Improper instrument selection
3. Misread a scale,
4. Forget a digit when reading a scale or recording a measurement.

- These blunder should stick out if we make multiple measurements or if one person checks the work of another.

Remedy:

  - Choose the best suitable instrument, based on the range of values to be measured.
  - Note down the readings carefully
- Blunders should not be included in the analysis of data.

Impacts of Errors on Accuracy and Precision

Impact	Accuracy	Precision
Systematic Errors	• Reduce accuracy but do not affect precision (results are consistently biased).	• Results remain consistent but incorrect.
Random Errors	• Reduce precision but may not significantly affect accuracy if repeated measurements are averaged.	• Results become inconsistent.
Gross Errors	• Affect both accuracy and precision significantly.	• Unreliable results.

Accuracy vs Precision

Systematic vs Random Errors

Volume (mL) of Cough Medicine Delivered by 10-oz (296 mL) Dispensers

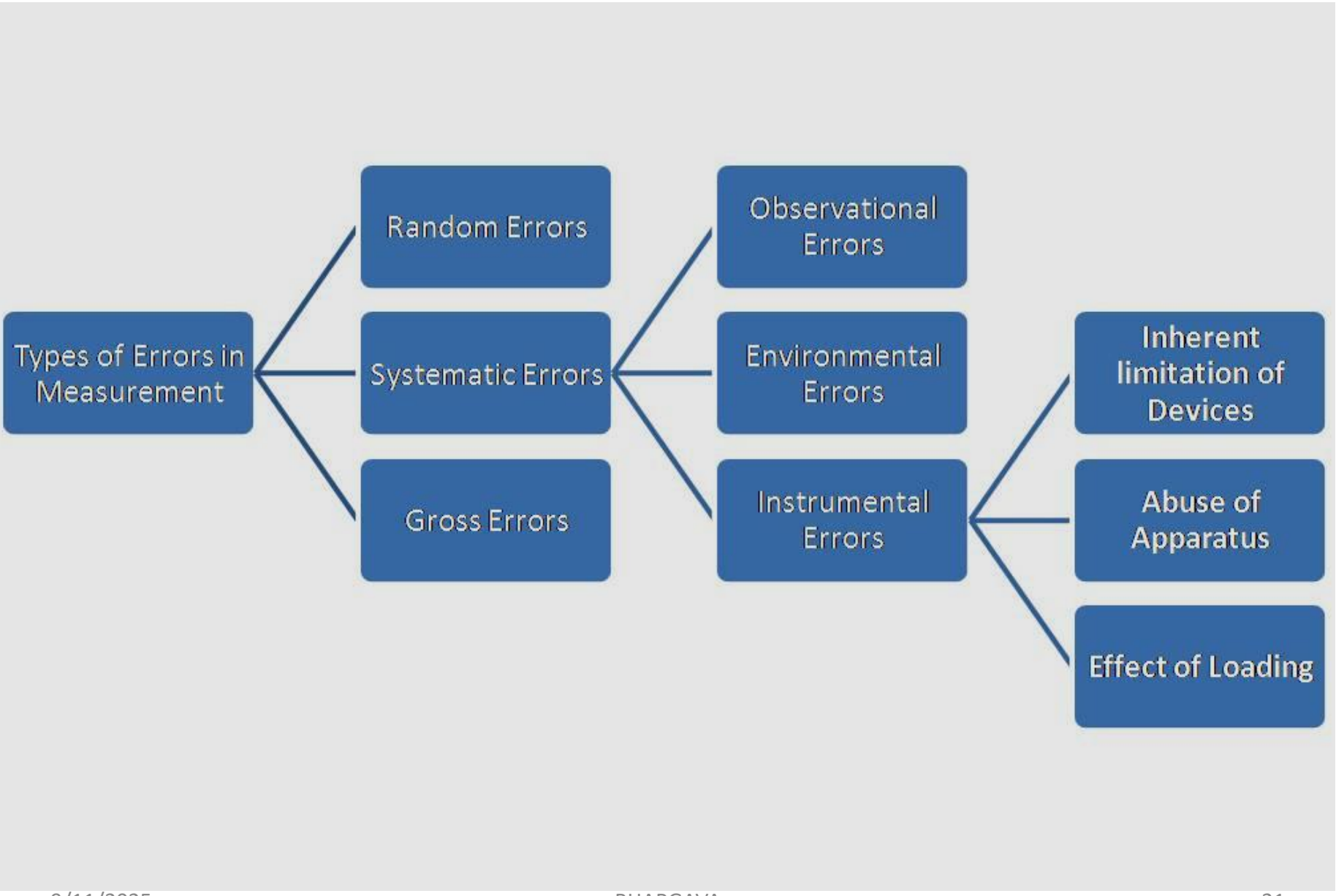
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284.1	293.9	296.1

Systematic Errors vs Random Errors

- Random errors lead to fluctuations around the true value as a result of difficulty taking measurements, while systematic errors lead to a predictable and consistent departure from the true value.
- Random errors are essentially unavoidable, while systematic errors are not.
- Random error varies unpredictably from one measurement to another, while systematic error has the same value or proportion for every measurement.
- Random errors are unpredictable and can’t be replicated by repeating the experiment again, while systematic errors produce consistent errors that can be replicated.
- Systematic error is a consistent, repeatable deviation from the true value in a measurement or experiment, often caused by faulty equipment or biased methodology, while random error is an unpredictable, inconsistent deviation from the true value due to unpredictable factors, such as measurement techniques or environmental conditions.

Systematic vs Random Errors

Feature	Systematic Errors	Random Errors
Nature	Predictable and consistent bias in one direction (positive or negative)	Unpredictable fluctuations, varying in both directions
Source	Arises from faulty equipment, poor calibration, or method errors	Arises from uncontrollable variables or inherent instrument noise
Effect on Results	Affects accuracy, causing measurements to deviate from the true value	Affects precision, causing scatter or variability in data
Correction	Can be corrected by calibration, improved design, or correction factors	Can be minimized by averaging multiple measurements or increasing precision
Example	A thermometer always reading 1°C higher than the actual temperature	Slight fluctuations in electronic noise when using a voltmeter
Detection	Systematic errors are detected by comparison with a known standard or reference	Random errors are observed as variability in repeated measurements


Types of Errors in Measurement
Random Errors
Systematic Errors
Gross Errors
Observational Errors
Environmental Errors
Instrumental Errors
Inherent limitation of Devices
Abuse of Apparatus
Effect of Loading

Uncertainty and Significant Figures

- Counted or defined numbers are always exact (without uncertainty).

- Measured values are always uncertain (level of uncertainty varies) because-
  - Due to practical limitations of instrument

- Significant figures (or significant digits) are the digits in a number that carry meaningful information about its precision. These include all non-zero digits, zeroes between significant digits, and trailing zeros in decimal numbers.

Role of Significant Figures in Calculations

- Significant figures play a crucial role in ensuring accuracy and precision during mathematical operations in scientific calculations, including forensic science applications.

- **Importance in Forensic Calculations**
  1. **Precision in Measurement:** Forensic analyses often involve measurements of weight, volume, concentration, and distances. Reporting these measurements with the correct number of significant figures ensures accuracy in evidence interpretation.
  2. **Chemical Analysis:** In forensic chemistry, calculations involving drug purity, explosive concentrations, and toxicology tests require careful attention to significant figures to maintain scientific integrity.
  3. **Error Minimization:** Rounding errors can affect forensic conclusions, especially in high-stakes cases like drug analysis or blood alcohol content determination.
  4. **Instrument Precision:** The number of significant figures reflects the precision of the measuring instrument used in forensic labs.
  5. **Courtroom Evidence:** Accurate and precise reporting of results increases the reliability of forensic evidence presented in court.

Types of ZERO



Rule-1

All non-zero digits are significant.

- **Explanation:** Any digit that is not zero counts as a significant figure.

- **Example:**
  - 234 has **three significant figures**.
  - 56.78 has **four significant figures**.

Rule-2

Zeros between non-zero digits are significant.

- **Explanation:** Any zero between two non-zero digits is counted as a significant figure.

- **Example:**
  - 105 has **three significant figures**.
  - 2007 has **four significant figures**.
  - 40.05 has **four significant figures**.

Rule-3

Leading zeros (zeros to the left of the first non-zero digit) are NOT significant.

- **Explanation:** Zeros that appear before the first non-zero digit are only placeholders and do not count as significant figures.

- **Example:**
  - 0.0035 has **two significant figures** (3 and 5; the zeros before 3 are not significant).
  - 0.00056 has **two significant figures** (5 and 6).

Rule-4

Trailing zeros in a number with a decimal point are significant.

- **Explanation:** Zeros at the end of a number **after** the decimal point are significant.

- **Example:**
  - 45.00 has **four significant figures**.
  - 0.0600 has **three significant figures** (6, and two trailing zeros after the decimal).
  - 2.000 has **four significant figures**.

Rule-5

Trailing zeros in a whole number without a decimal point are NOT significant.

- **Explanation:** Zeros at the end of a number **without** a decimal point are just placeholders and are not considered significant.

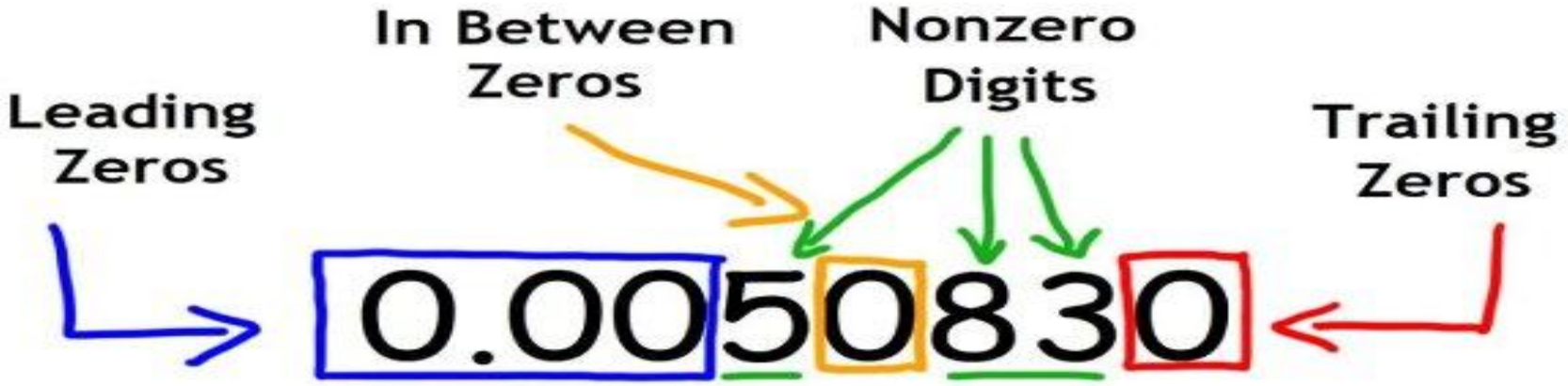
- **Example:**
  - 2500 has **two significant figures** (2 and 5; the trailing zeros are not significant).
  - 500 has **one significant figure** (5; the two zeros are not significant).
  - 3000 has **one significant figure** (3).

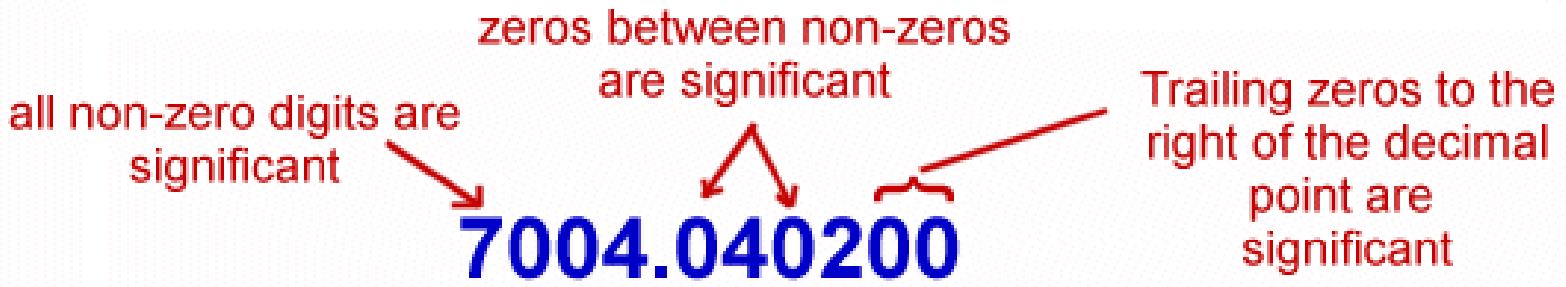
Rule-6

In scientific notation, all digits in the coefficient are significant.

- **Explanation:** In numbers expressed in scientific notation, all the digits in the coefficient (the number part before the "×10") are considered significant.

- **Example:**
  - 4.50×10^{3} has **three significant figures** (4, 5, and 0).
  - 6.70×10^{-2} has **three significant figures** (6, 7, and 0).
  - 1.0×10^{4} has **two significant figures** (1 and 0).





These zeros are not significant digits

3,200 **0.004709**

Let's look at some examples



a. 508.0 L — 4 significant figures

b. 820 400.0 L — 7 significant figures

c. 1.0200×10^5 kg — 5 significant figures

d. 807 000 kg — 3 significant figures



Examples

- 10082 —— 5 sig figs
- 70,000 —— 1 sig fig
- 0.0025 —— 2 sig figs
- 0.000309 —— 3 sig figs
- 50010.000 —— 8 sig figs
- 0.0040030 —— 5 sig figs

Significant Figures during Rounding

- If the digit to be dropped (the one immediately to the right of the digit to be retained) is **less than 5**:
  - We “round down” and leave the retained digit unchanged.

- If the dropped digit is **more than 5**:
  - We “round up” and increase the retained digit by 1.

- If the dropped **digit is 5**:
  - We round up or down, whichever yields an even value for the retained digit.

- 0.028675 rounds “up” to 0.0287 (the dropped digit, 7, is greater than 5)
- 18.3384 rounds “down” to 18.3 (the dropped digit, 3, is lesser than 5)
- 6.8752 rounds “up” to 6.88 (the dropped digit is 5, and the retained digit is even)
- 92.85 rounds “down” to 92.8 (the dropped digit is 5, and the retained digit is even)

Practice problems (rounding)

- Round the following to the indicated number of significant figures:

  1. 31.57 (to two significant figures)
  2. 8.1649 (to three significant figures)
  3. 0.051065 (to four significant figures)
  4. 0.90275 (to four significant figures)

1. 31.57 rounds “up” to **32** (the dropped digit is 5, and the retained digit is even)
2. 8.1649 rounds “down” to **8.16** (the dropped digit, 4, is lesser than 5)
3. 0.051065 rounds “down” to **0.05106** (the dropped digit is 5, and the retained digit is even)
4. 0.90275 rounds “up” to **0.9028** (the dropped digit is 5, and the retained digit is even)

Significant Figures during Calculations

- When we add or subtract numbers:
  - We should round the result to the same number of decimal places as the number with the least number of decimal places (the least precise value in terms of addition and subtraction).

a) Add 1.0023 g and 4.383 g.

b) Subtract 421.23 g from 486 g

(a)
$$\begin{array}{r} 1.0023\ \text{g} \\ +\ 4.383\ \text{g} \\ \hline 5.3853\ \text{g} \end{array}$$

Answer is 5.385 g (round to the thousandths place; three decimal places)

(b)
$$\begin{array}{r} 486\ \text{g} \\ -421.23\ \text{g} \\ \hline 64.77\ \text{g} \end{array}$$

Answer is 65 g (round to the ones place; no decimal places)

Practice problems

(a) Add 2.334 mL and 0.31 mL.

(b) Subtract 55.8752 m from 56.533 m.

Answer:

(a) 2.64 mL

(b) 0.658 m

Significant Figures during Calculations

- When we multiply or divide numbers:
  - We should round the result to the same number of digits as the number with the least number of significant figures (the least precise value in terms of multiplication and division).

(a) Multiply 0.6238 cm by 6.6 cm.

(b) Divide 421.23 g by 486 mL.

(a) $0.6238\ \text{cm} \times 6.6\ \text{cm} = 4.11708\ \text{cm}^2 \longrightarrow$ result is $4.1\ \text{cm}^2$ (round to two significant figures)

four significant figures × two significant figures ⟶ two significant figures answer

(b) $\dfrac{421.23\ \text{g}}{486\ \text{mL}} = 0.86728...\ \text{g/mL} \longrightarrow$ result is $0.867\ \text{g/mL}$ (round to three significant figures)

$\dfrac{\text{five significant figures}}{\text{three significant figures}} \longrightarrow$ three significant figures answer

Practice problems

(a) **Multiply** 2.334 cm and 0.320 cm.

(b) **Divide** 55.8752 m by 56.53 s.

Answer:

(a) 0.747 cm^2

(b) 0.9884 m/s

Practice problem

- A forensic chemist is analyzing a sample for drug content. The following measurements are recorded: 25.345 grams of the sample and a purity concentration of 92.3%. Calculate the total mass of the drug in the sample while maintaining the correct significant figures. Explain your steps.

- Sample mass = **25.345 grams**
- Purity concentration = **92.3%**

- **Total Drug Mass**= Sample Mass × $\frac{\text{Purity Percentage}}{100}$

  =25.345 × 92.3/100

  =25.345 × 0.923

  =23.388935 grams

  = 23.4 grams